Lab 6: Medium Access Control: Aloha and CSMA

In this lab you simulate a single channel shared by many uncoordinated stations and measure how much useful traffic it carries as the offered load rises. You implement three access methods (pure Aloha, slotted Aloha, and 1-persistent CSMA), plot the throughput of each against offered load, and compare your curves to the analytical results from class. The medium is the same-shared channel as before, but now many senders use it at once, and the new failure is the collision.

This lab is statistical, not deterministic. You do not match an exact output; you run a Monte Carlo simulation and confirm that your measured curve lies on the analytical curve and that its peak lands at the right load.

A Monte Carlo simulation estimates a quantity by generating many random trials and averaging the outcome, instead of computing it from a formula. Here you generate random frame arrivals over many frame-times and measure the fraction that succeed; that measured fraction is your estimate of throughput.

You will:

- simulate pure Aloha and reproduce its curve, peaking near S = 0.18 at G = 0.5;
- simulate slotted Aloha and reproduce its curve, peaking near S = 0.37 at G = 1;
- simulate 1-persistent CSMA, where a station senses the channel before transmitting, and show its throughput rises well above both;
- Explain from your own data why slotting doubles pure Aloha and why sensing beats both.

Setup

Use any C++ compiler or Python 3. Publish and use a fixed random seed (use 42) and simulate at least H = 10000 frame-times, so your results are reproducible and stable. Vary G from 0.1 to 3.0 and record the throughput at each.

The model you build

One channel, one arrival stream, three access rules. These remain constant:

- Time is measured in frame-times. Every frame occupies exactly one frame-time.
- Offered load G is the number of transmission attempts per frame-time. G already includes both new frames and retransmissions, so you generate the aggregate load directly and do not simulate backoff.
- Arrivals are one aggregate Poisson stream at rate G, not one stream per station. Over H frame-times, draw a Poisson number of attempts with mean G times H and place them uniformly in time (for slotted Aloha, equivalently draw a Poisson count per slot with mean G).
- Throughput S is the number of successful transmissions divided by H.

The three protocols differ only in the access-and-collision rule, which is the size of the vulnerable window.

Part A: Pure Aloha

What you do: simulate stations that transmit the moment a frame arrives, detect collisions, and measure throughput against offered load.

Background: a frame that starts at time t occupies the channel until t plus one frame-time. It succeeds only if no other frame starts within one frame-time before it or one frame-time after it. That two-frame-time span is the vulnerable window.

Your task:

1. Generate the aggregate arrivals for a given G.
2. Mark a frame successful if no other arrival lies within one frame-time on either side of it.
3. Measure S and sweep G from 0.1 to 3.0.
4. Plot S against G with the analytical curve S = G e^(-2G) overlaid, and confirm the peak of about 0.184 at G = 0.5.

Part B: Slotted Aloha

What you do: divide time into slots of one frame-time, allow transmissions to begin only at slot boundaries, and measure throughput.

Background: because every frame now begins at a slot boundary, partial overlap disappears. A slot succeeds only if exactly one station transmits in it, so the vulnerable window shrinks from two frame-times to one.

Your task:

1. For each slot, determine how many stations transmit (a Poisson count with mean G).
2. Mark a slot successful if exactly one station transmits.
3. Measure S and sweep G.
4. Plot S against G with the analytical curve S = G e^(-G) overlaid, and confirm the peak of about 0.368 at G = 1. State, from your two curves, why slotting exactly doubles the peak.

Part C: CSMA (1-persistent)

What you do: make a station sense the channel before transmitting, defer while it is busy, and transmit as soon as it is idle. Show that throughput climbs far above slotted Aloha.

Background: before transmitting, a station listens. If the channel is idle it transmits; if busy it waits and transmits the instant the channel frees. Collisions do not disappear, because sensing is not instantaneous. A transmission that starts at time s only appears busy to others after a

propagation delay a, so any station that starts within a of s also sensed idle and collides. Deferred stations released together when the channel frees collide for the same reason. Use a small delay ratio, a = 0.01.

Your task:

1. Implement carrier sensing: an arrival transmits if the channel is idle, otherwise it defers.
2. Implement the release: deferred arrivals transmit when the channel next becomes idle.
3. Mark a transmission successful only if no other transmission starts within one propagation delay a of it.
4. Sweep G, plot S against G, and report the peak throughput you measure and the load at which it occurs. It should be near 0.53 at G = 1 and clearly above slotted Aloha.

Part D: Comparison and analysis

What you do: put the three curves on one axis and account for their differences from your own data.

Your task:

1. Overlay the three throughput curves.
2. Explain why the Aloha curves collapse at high load while the CSMA curve does not.
3. Identify the single quantity that controls how close CSMA comes to full utilisation, and, by rerunning Part C with a = 0.05 and a = 0.1, show what happens to the CSMA peak as that quantity grows.
4. State the vulnerable window of each protocol (two frame-times, one frame-time, one propagation delay) and connect it to the height of each peak.

What to submit

- Your simulation source (C++ or Python).
- The three throughput curves, with the analytical curve overlaid for Parts A and B, and the measured peak of each protocol.
- The written answers to Part D, including the CSMA peaks for a = 0.01, 0.05, and 0.1.